

Logbook: validation of `proVBFH`, `proVBFHH` and `proVBFH-inclusive` against the published results, the HH tensor matrix element, the non-factorisable corrections, and CI

24–25 September 2026

This logbook summarises the work recorded in `README.md` in the same directory, which remains the detailed record (with the file-by-file description of `runs/`, `scripts/` and `patches/`). All work is on the branch `ci`, pushed to GitHub; commits are quoted by their short hashes (list in appendix A).

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1 Setup

The goal was to set up continuous integration (CI), and before that to check that the current code, built against the latest dependency releases, reproduces the published numbers:

- 1506.02660 (VBF H at NNLO, fully differential),
- 1606.00840 (VBF H at N^3 LO, inclusive),
- 1811.07906 (VBF HH at N^3 LO, inclusive),
- 1811.07918 (VBF HH at NNLO, fully differential),

- 2005.11334 (non-factorisable corrections to VBF H and HH).

The PDFs of the papers are in `papers/`. Dependencies: `hoppet` 2.3.0, LHAPDF 6.5.6, FastJet 3.5.1, gfortran 15.2 (Linux), i.e. the latest releases used by the CI.

2 Inclusive cross sections (proVBFH-inclusive)

Common flags: `-ncall1 1000000 -ncall2 4000000 -itmx1 3 -itmx2 3 -7scaleuncert`, one run per order. MC errors are ± 0.0013 pb (H) and ± 0.0011 fb (HH).

order	H, 13 TeV, 1606.00840 [pb]		HH, 14 TeV, 1811.07906 [fb]	
	this run	paper	this run	paper
LO	$4.0962^{+0.0543}_{-0.0643}$	$4.099^{+0.051}_{-0.067}$	$2.0749^{+0.1800}_{-0.1489}$	$2.079^{+0.177}_{-0.152}$
NLO	$3.9677^{+0.0274}_{-0.0201}$	$3.970^{+0.025}_{-0.023}$	$2.0616^{+0.0216}_{-0.0147}$	$2.065^{+0.022}_{-0.018}$
NNLO	$3.9294^{+0.0178}_{-0.0079}$	$3.932^{+0.015}_{-0.010}$	$2.0519^{+0.0046}_{-0.0025}$	$2.056^{+0.003}_{-0.005}$
N ³ LO	$3.9270^{+0.0071}_{-0.0000}$	$3.928^{+0.005}_{-0.001}$	$2.0514^{+0.0034}_{-0.0000}$	$2.055^{+0.001}_{-0.001}$

Table 1: Inclusive cross sections, $\mu = Q_i$. H: PDF4LHC15_nnlo_mc, $M_H = 125.09$, $M_W = 80.385$; HH: PDF4LHC15_nnlo_mc, $M_H = 125$, $M_W = 80.379$, $\Gamma_H = 4.030 \cdot 10^{-3}$. HH at 27 TeV, N³LO: 8.3945 ± 0.0050 vs 8.407 fb.

The no-cuts columns of the differential papers ($\mu_0(p_t)$ scale) agree similarly. H (NNPDF30), LO and NLO: 4.0276 and 3.9250 pb vs 4.032 and 3.929 pb; HH (PDF4LHC15): 2.0135 and 2.0470 fb vs 2.016 and 2.049 fb.

The small offset. All inclusive numbers came out slightly low (H by 0.03–0.1%, HH by about 0.18%), by the same amount at every order including LO, so the cause is in the inputs rather than in the QCD corrections (halving the `hoppet` y grid spacing changes HH LO by $2 \cdot 10^{-6}$). Three changes since the papers account for it (LO, same seed, cumulative):

change relative to the paper-era code	shift of HH LO
$Q_{\min}$: $\sqrt{2} \rightarrow 1$ GeV (the cut raised in <code>bfdb73e</code> , 2025)	+0.051%
EW masses passed to <code>hoppet</code> 's <code>StartStrFct</code> ($\sin^2 \theta_W$ of the Z couplings)	+0.038%
old tensor matrix element (the 2.0.0 default) instead of the analytic one	+0.023%

For H, $Q_{\min}$ alone accounts for about half of the offset (+0.037% at $\mu = Q$, +0.040% at $\mu_0(p_{t,H})$). Averaged over four further seeds, HH LO is 2.07745 ± 0.00054 fb with the current code and 2.07983 ± 0.00055 fb in the paper-era configuration ($Q_{\min} = 1$, 2018 EW defaults of `hoppet`, old tensor ME), vs 2.079 in the paper: the paper-era configuration reproduces 1811.07906. The EW-mass change is deliberate (the structure functions and the matrix element now use the same M_W).

3 Differential cross sections with VBF cuts (proVBFH, proVBFHH)

From `example/` with `runningscales 1`, the papers' PDFs and EW parameters (H: 13 TeV, NNPDF30_nnlo_as_0118; HH: 14 TeV, PDF4LHC15_nnlo_mc). LO: 8 seeds with 10^7 calls; NLO: 8 cores with `aux/runpar.sh`.

process, order	this run	paper
H LO	0.95722 ± 0.00025 pb	0.957 pb
H NLO	0.8740 ± 0.0014 pb	0.876 pb
HH LO	0.79879 ± 0.00018 fb	0.799 fb
HH NLO	0.7310 ± 0.0024 fb	0.726 fb

Bug found (bc33959). Since 828ff81 (March 2025) the exclusive part of `proVBFHH` read `cVHHfact`, `cVHfact` and `lambdafact` without a default; when absent from `powheg.input` (as in `example/`) they were -10^6 , and all HH NLO/NNLO runs gave about 10^{20} fb. LO and inclusive runs were unaffected. The smoke tests alone would not have caught this.

4 The HH tensor matrix element

4.1 The two implementations

The HH matrix element contracts the hadronic tensors of the two beams with the $VV \rightarrow HH$ current,

$$\sigma \propto W_{\mu\nu}^{(1)} W_{\rho\sigma}^{(2)} M^{\mu\rho} M^{*\nu\sigma}, \quad W_{\mu\nu} = F_1 G_{\mu\nu}^{(1)} + F_2 G_{\mu\nu}^{(2)} + F_3 G_{\mu\nu}^{(3)}, \quad (1)$$

with the basis tensors

$$G_{\mu\nu}^{(1)} = \frac{q_\mu q_\nu}{q^2} - g_{\mu\nu}, \quad G_{\mu\nu}^{(2)} = \frac{\hat{P}_\mu \hat{P}_\nu}{P \cdot q}, \quad G_{\mu\nu}^{(3)} = \frac{i \epsilon_{\mu\nu\rho\sigma} P^\rho q^\sigma}{2 P \cdot q}, \quad \hat{P} = P - \frac{P \cdot q}{q^2} q, \quad (2)$$

and the current

$$M^{\mu\nu} = A g^{\mu\nu} + B (2k_1 + q_1)^\mu (k_2 - k_1 - q_1)^\nu + C (2k_2 + q_1)^\mu (k_1 - k_2 - q_1)^\nu, \quad (3)$$

where A collects the triangle (s -channel Higgs), the quartic vertex and the $g^{\mu\nu}$ part of the t/u -channel boxes, and B, C the remaining t/u -channel structures (with the coupling modifiers c_V, c_{VV}, λ). The default `eval_matrix_element` uses the analytic contraction (Mathematica expressions in `ME_expressions.f`); `eval_matrix_element_tensor` (`-tensorME`, `tensorME 1`) contracts numerically. The tensor version is the only one with the diagram switches, and the non-factorisable (NF) corrections always use it.

4.2 The F_3 bug was in the tensor version

A first analysis (same-point comparison of the two implementations) took the tensor version as the reference and concluded that the $F_3 \times F_3$ interference terms of the analytic **F3F3** were wrong. **That was backwards.** The tensor code stored the contravariant components of q, P and k_i in tensors declared with lower indices, i.e. it worked with the parity-flipped vectors, but built $\epsilon_{\mu\nu\rho\sigma} P^\rho q^\sigma$ from the unflipped ones. Everything without ϵ is parity invariant and came out right, as did $F_3 \times F_3$ for A alone; the $F_3 \times F_3$ interference of A with B, C did not.

An independent calculation with explicit index handling at a physical phase-space point (`scripts/f3-check/`) settles it: the analytic **F3F3** coefficients (AA, AB, AC, BC) agree with it to all printed digits, the old tensor convention only for AA (AB is off by a factor 39, AC by 4.4, BC by 22). Fixed in 03adc62: vectors are now lowered with the metric (`InitFourVector` in `tensor.f90`). The effect on the integrated HH cross section is at the 10^{-4} level (analytic/old tensor $-1 = (-1.5 \pm 0.9) \cdot 10^{-4}$).

4.3 What remains: parity-odd width terms

With finite widths the tensor version contains $F_1 F_3$ and $F_2 F_3$ terms, $\text{Im}(AB^*)$ etc. times the $i\epsilon$ part of W , which the analytic version does not have. Each has a single ϵ tensor, so they are parity

odd and integrate to zero for σ and every parity-even distribution. Same-point comparisons after the fixes (`scripts/dbg_patch.py`, 0.6M points, HH at 14 TeV):

comparison	analytic/tensor -1 (integral)	pointwise
zero widths, LO	$(-2 \pm 9) \cdot 10^{-16}$	$\leq 1.4 \cdot 10^{-5}$ (rare points)
zero widths, N ³ LO	$(+4 \pm 9) \cdot 10^{-16}$	$\leq 5.9 \cdot 10^{-6}$
physical widths, LO	$(-2.2 \pm 1.2) \cdot 10^{-4}$	up to 46% of the scale
widths, mirror-averaged tensor, LO	$+7 \cdot 10^{-16}$	$\leq 1.4 \cdot 10^{-9}$ of the scale
same, N ³ LO	$-1 \cdot 10^{-15}$	$\leq 3.9 \cdot 10^{-9}$ of the scale

(The scale is the sum of the absolute values of all terms; “mirror-averaged” means the tensor matrix element averaged over each point and its mirror image $y \rightarrow -y$, which removes its parity-odd part.) So the analytic version is correct for all parity-even observables. **Decision: the analytic matrix element stays the default**; the tensor one is needed for parity-odd observables, the diagram switches and the NF corrections.

4.4 Order combination

The analytic version expands strictly, $\sum_n \sum_{i+j=n+1} F_{\text{beam } 1}^{(i)} F_{\text{beam } 2}^{(j)}$. The old tensor version summed the structure functions over all orders first, which adds products beyond the working order and ignores `order_start`: non-tensor/old tensor -1 without F_3 was $2 \cdot 10^{-15}$ (LO), $+2.7 \cdot 10^{-4}$ (NLO), $+0.65 \cdot 10^{-4}$ (NNLO) and $+0.06 \cdot 10^{-4}$ (N³LO). Since W is linear in the structure functions, the new version (`edc7a24`) computes once per point and boson

$$T_{kl} = \text{Re} \left[G_{1,\mu\nu}^{(k)} G_{2,\rho\sigma}^{(l)} M^{\mu\rho} M^{*\nu\sigma} \right], \quad \sigma \propto \sum_n \sum_{i+j=n+1} \sum_{k,l} F_k^{(i)} F_l^{(j)} T_{kl}, \quad (4)$$

exactly the strict expansion, at no extra cost per order. With zero widths, same-seed runs with and without `-tensorME` now give identical results and MC errors at every order, also with coupling modifiers (which the `proVBFH-inclusive` tensor version ignored before `44d0d79`).

4.5 Code changes and performance

- `8d00b5d`: `tensor.f90` rewritten without allocatable components (bit-identical results), the same file in all three packages. The previous version is kept as the oracle in `proVBFH-inclusive` (`tests/`); `make check` runs a unit test against it and against explicit formulae (tolerances robust to fused multiply-adds since `f706237`).
- `af68ff5`: `proVBFH` used `cmplx()` without `kind=dp`, i.e. single-precision propagators ($1.5 \cdot 10^{-8}$ effect).
- Timing of `proVBFH_incl` (1M points, one core): old tensor 7.4 s (LO) / 9.1 s (N³LO), new tensor 4.4 / 6.1 s, analytic 2.7 / 4.4 s.

5 Non-factorisable corrections

5.1 Validation against 2005.11334

Setup of 1506.02660 (13 TeV, NNPDF30, $M_W = 80.398$, $\Gamma_W = 2.141$), `qcd_order 1`, `nonfact 1`, central scale only. H: $\delta\sigma_{\text{NF}} = -0.0030251 \pm 0.0000098$ pb vs -0.0030 pb (-0.316% of LO vs -0.32%). HH, table 2 of the paper, with the fixed tensor code (paper in brackets):

	σ_{TT}	σ_{BB}	σ_{TB}	Σ
Born [fb]	10.368 (10.393)	14.128 (14.172)	-23.838 (-23.904)	0.6577 (0.662)
1-loop NF	0.342% (0.339%)	0.303% (0.300%)	0.320% (0.318%)	0.283% (0.286%)
2-loop NF	-0.666% (-0.667%)	-0.620% (-0.621%)	-0.643% (-0.644%)	-0.516% (-0.516%)
full NF	-0.324% (-0.327%)	-0.318% (-0.320%)	-0.323% (-0.326%)	-0.233% (-0.230%)

MC errors: 0.2–0.5% (relative) for TT, BB; for the Σ column, which has a strong cancellation, ± 0.010 , ± 0.004 and ± 0.011 (absolute, in %). With the old tensor code the Σ column was 0.274%, -0.513% , -0.239% (compatible within errors); the Born TT result is unchanged to the last digit, as it must be (triangles have only the $g^{\mu\nu}$ structure, so the $A \times B$ bug cannot affect them). The Born is 0.25–0.3% below the paper with NNP30 and about as much above with PDF4LHC15; the paper does not state its HH PDF. This does not affect the NF percentages and was accepted as is.

5.2 Speed-up of the angular integrals

Each phase-space point needs integrals over the gluon azimuth $\xi \in [0, 2\pi]$ of the eikonal integrands of `nonfact_expressions.f` (HH: 8 per boson, H: 2). They were done with a Runge–Kutta routine with `niter` = 100 steps; since the integrands do not depend on y , this is Simpson’s rule with the midpoint and the left endpoint evaluated twice, i.e. 400 calls per integral, half of them redundant.

A study on 9000 argument sets dumped from real HH and H NF runs (in `scripts/`) showed that the integrands are smooth for typical kinematics, but at large transverse momenta develop peaks of width $\sim M_V/p_T$ where ξ aligns with an external transverse momentum. No fixed rule copes (maximum error relative to $\int |f|$):

rule (calls)	b_{01}	b_{022}	t_{022}
RK4, <code>niter</code> 100 (400)	$1.9 \cdot 10^{-4}$	$8.1 \cdot 10^{-5}$	$1.6 \cdot 10^{-1}$
trapezoidal (64)	$2.5 \cdot 10^{-3}$	$1.1 \cdot 10^{-3}$	1.1
Gauss–Legendre (128)	$4.0 \cdot 10^{-5}$	$1.8 \cdot 10^{-5}$	$8.5 \cdot 10^{-3}$
adaptive G–K, $\epsilon = 10^{-8}$ (mean calls)	$1.0 \cdot 10^{-11}$ (82)	$9 \cdot 10^{-10}$ (77)	$1.0 \cdot 10^{-9}$ (114)

For RK4 on t_{022} the median error is $2 \cdot 10^{-15}$, the 99th percentile $2 \cdot 10^{-3}$. The changes:

- **f37f0da**: the roots $r_1 \dots r_6$ were single-precision complex (`complex*8`; effect at most $1.6 \cdot 10^{-4}$ on the integrals).
- **d593bb4**: adaptive Gauss–Kronrod (G7/K15, 4 initial intervals, bisection of the worst interval) to a relative accuracy `nf_epsrel` (new input, default 10^{-8}) of $\int |f|$; `niter` is no longer used.
- The integrands have the structure (with $L_k = \log(-r_k/M_V)$, $D_k = \pi r_k \prod_{j \neq k} (r_k - r_j)$)

$$b_{01} = - \sum_{k=1,3,5} \frac{L_k}{D_k}, \quad b_{022} = -2 \sum_{k=1,3,5} \frac{L_k^2}{D_k}, \quad b_{12} = -2 b_{01}, \quad (5)$$

and likewise $t_{01}, t_{022}, t_{12} = -2t_{01}$ for the triangle with $k = 1, 3$ and $j \leq 4$. **03a4587** integrates the 1- and 2-loop boxes together (shared roots, logs and peaks), and **1a53e5e** folds the triangle into the integration of the first box of each boson (the box denominators are the triangle ones times $(r_k - r_5)(r_k - r_6)$).

Full HH NF run (notes statistics, 8 runs in parallel on 12 cores): 2332 s $\rightarrow$ 482 s with the adaptive integration, 437 s with the folding, i.e. about $5\times$ faster. Every NF result (HH table, H) is identical to the old one in every printed digit.

Impact on distributions. The old integration was inaccurate only at rare, extreme kinematics. Comparing all histograms of the analysis bin by bin between the old and new runs (same seeds, otherwise the same code; 35 histograms for HH, 14 for H), the largest shift is 1.3% in the $m_{HH} = 3.2\text{--}3.4$ TeV bin of the 1-loop HH run, 5% of its MC error; for the full HH NF correction 0.4% ($p_{T,H} = 780\text{--}810$ GeV, 1.4% of the MC error); for H $2 \cdot 10^{-5}$. The paper’s figures show NF/LO as a function of m_{jj} and Δy_{jj} selection cuts: the cross sections above an m_{jj} or Δy_{jj} cut change by at most 10^{-5} (full HH NF), $1.4 \cdot 10^{-4}$ (1-loop HH, $m_{jj} > 6.8$ TeV) and $2 \cdot 10^{-8}$ (H). The 2D cut grid of the figures was not rebuilt.

5.3 Regulator independence

The gluon-mass regulator was hard-wired to $\lambda = M_V^2$, where $L \equiv \log(\lambda/M_V^2) = 0$. With the identities above, the 1- and 2-loop functions have the form

$$\chi_a^{(1)} = f_a - L, \quad \chi_a^{(2)} = C_a - 2f_a L + L^2, \quad (6)$$

for each diagram class a , with $M_1 = i \sum_a \chi_a^{(1)} X_a$ and $M_2 = -\frac{1}{2} \sum_a \chi_a^{(2)} X_a$, so that the L terms cancel point by point in $|M_1|^2 + 2 \operatorname{Re}(M_2 M_0^*)$. `fe67abd` therefore obtains b_{12}, t_{12} from the 1-loop integrals and adds the input `nf_regfact` ($\lambda = \text{nf_regfact} \cdot M_V^2$, default 1). The 1-loop triangle is computed analytically (`chi_tri1`); the numerical $-2 \int t_{01}/t_{11}$ agrees with it to $1.7 \cdot 10^{-11}$, so λ independence also tests the angular integration. Same-seed runs with `nf_regfact` 0.01, 1 and 100 give identical results in all printed digits (HH all diagrams, HH triangles only, H).

6 Continuous integration

`.github/workflows/ci.yml` builds the three packages on `ubuntu-latest` and `macos-latest` against the latest releases of `hoppet`, `LHAPDF` and `FastJet` (on push/PR, weekly, on demand), with the logic in `ci/*.sh` so that it runs locally too:

- `smoke-tests.sh`: short runs of every program, checking for complete and finite output (not a physics validation);
- `tensor-checks.sh`: `make check`, identical copies of `tensor.f90`, and tensor vs analytic matrix element at zero widths (LO and NNLO);
- `nf-checks.sh`: NF runs of `proVBFH` and `proVBFHH` with $\lambda = M_V^2$ and $0.01 M_V^2$, which must agree to 10^{-6} (a deliberately wrong sign in $b_{12} = -2b_{01}$ changes the result by a factor 100);
- `annotate.sh`: reports the error lines of a failing step as annotations in the run summary.

The first macOS runs exposed: the unit test demanded bit-identical rounding (fails with FMA on arm64); the tensor module stopped with exit status 0 on errors (now `error stop`); the Fortran module-directory flag reached clang++; `-mcmmodel=large` is not supported there (`f706237, b70d3af`). CI is green on both platforms. The dependency cache is saved right after installing (`b8f1eb2, 91f8bbb`; the saved caches, 25 MB on macOS and 39 MB on Linux, are now in place). Also fixed on the way: `make -j` failed (`44e051a`).

7 Open points

- HH Born normalisation vs 2005.11334 (PDF not stated in the paper): accepted as is.
- The 2018 HH papers used the old tensor code: their numbers include the F_3 error and the beyond-accuracy order products, both at the 10^{-4} level, far below the quoted uncertainties.
- Possible further NF speed-up: initial Gauss–Kronrod break points at the known peak angles.
- `incl2pwhg.f` ignores `tensorME` (consistent with keeping the analytic default).
- Merging `ci` into `main`.

A Commits on the ci branch

bc33959 `proVBFHH`: default coupling modifiers to 1 in the exclusive code
1326b19 `proVBFH-inclusive`: set `rpath` for `hoppet` and `LHAPDF`
44e051a `proVBFH`, `proVBFHH`: add missing `constants.o` dependency on `helper.o`
8d00b5d `tensor.f90`: allocation-free rewrite, shared by all three packages
03adc62 HH tensor matrix element: fix the F_3 interference terms
af68ff5 `proVBFHH`: double-precision `cmplx` in the matrix elements
edc7a24 HH tensor matrix element: combine orders per order pair
44d0d79 `proVbfhh_incl`: coupling modifiers in the tensor matrix element
caf5177 Add continuous integration (GitHub Actions)
8726b60 Add `CLAUDE.md`
7a2eafa notes: validation of the current code against the published results
f706237 tensor tests: robust to FMA; **error stop** in the tensor code
eb4bc86 CI: report errors as annotations; fix the artifact path
b70d3af `proVBFH`, `proVBFHH`: fix the build on macOS
f37f0da NF corrections: double-precision complex in `nonfact_expressions.f`
d593bb4 NF corrections: adaptive Gauss–Kronrod for the azimuthal integrals
03a4587 NF corrections (HH): integrate the 1- and 2-loop boxes together
b8f1eb2 CI: save the dependency cache right after installing
d6d42f0 notes: NF speed-up study and revalidation
fe67abd NF corrections: regulator input `nf_regfact`; b_{12} , t_{12} from the 1-loop integrals
1a53e5e NF corrections (HH): fold the triangle into the box integration
9f1d7c0 CI: check that the NF corrections do not depend on the regulator
9e08c09 notes: NF regulator independence and triangle folding
91f8bbb CI: fix the dependency cache; current action versions